

Internship Progress Report

Chronic Disease Burden & Social Determinants Dashboard

Date: March 19, 2026
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Organization: Global AI
Period: Week 1 — Part-Time Internship

Work Completed

My first week was mostly getting the data and understanding what I was actually working with before writing any real analysis code.

The CDC publishes their PLACES county health data and Social Vulnerability Index through public APIs, so I wrote a script to pull all six PLACES release years (2020–2025) and both SVI snapshots automatically into a local folder. That part went smoothly.

After that I spent a lot longer than expected just exploring the files. I compared columns across all six PLACES files, checked for missing values, and tried to figure out how the county IDs matched between the two datasets. I thought this would take maybe an hour — it ended up taking most of the week.

Data Acquisition

The CDC PLACES dataset and Social Vulnerability Index are both available through public Socrata APIs. I downloaded all six PLACES county-level release years (2020–2025), totalling approximately 1.25 million rows across six files, alongside both available SVI snapshots (2020 and 2022), each covering 3,100+ counties across 158 vulnerability indicators.

Exploratory Data Analysis

The first thing that confused me was that the filename year and the actual data year are different. “PLACES 2025” turns out to contain data from 2022 and 2023, not 2025. I did not catch this at first and had to go back to the CDC documentation to understand why. Once I figured it out, I added a check that reads the year column from inside each file so there is no guessing.

The trickier issue was the county FIPS codes. Both datasets store them as integers, which seems fine — until you realise counties in states like Alabama or Alaska have FIPS codes that start with a zero (e.g., 01001). An integer drops that leading zero and becomes 1001. When I tried joining the two datasets, the match count was lower than it should have been, and that is how I found it. The actual fix was one line of code, but tracking down why counties were going missing took a while.

A few other things worth noting: the 2020 PLACES file has no county ID column at all — just latitude and longitude — so I had to leave it out since I could not match it to the SVI data without a separate lookup table. The SVI dataset also uses -999 to mean “data unavailable” instead of leaving the field blank, which would have broken any calculations if I had not replaced those with proper null values.

Difficulties Encountered

Honestly, the data exploration part was harder than I expected. Real public health datasets are messier than what I have worked with in class. The PLACES program has added and dropped measures over the years, so not every county has every variable depending on the year — figuring out what was actually consistent and usable required a lot of back-and-forth.

A summary of the key issues identified and resolved:

- **Filename year $\neq$ data year:** Each PLACES file contains data from two earlier calendar years, not the release year shown in the filename.
- **FIPS zero-padding:** Both datasets store county FIPS codes as integers, silently dropping leading zeros and causing join mismatches for certain states.
- **Missing county identifier in 2020 file:** `places_2020` stores location as coordinates rather than a FIPS code and was excluded from the analysis.
- **SVI sentinel values:** The SVI dataset encodes missing data as `-999` rather than null, requiring explicit replacement before any analysis.
- **Measure availability varies by year:** Several SDOH measures (e.g., `FOODINSECU`, `HOUSINSECU`) were only added to the PLACES program in 2024 and are unavailable in earlier years.

Planned Work

I am planning to finish the data cleaning and merging over the weekend. That means pivoting the PLACES data from its current format (one row per county per health measure, approximately 200,000 rows) into a wider format (one row per county, one column per measure), then joining it to the SVI data on the FIPS code. Once I have a clean combined dataset I will save it out as its own file so I am not re-running everything from scratch each time.

Next week I will explore and finish the first task of the assignment.